

Disease Prediction using Symptoms

COEN 691 – Applied Machine Learning

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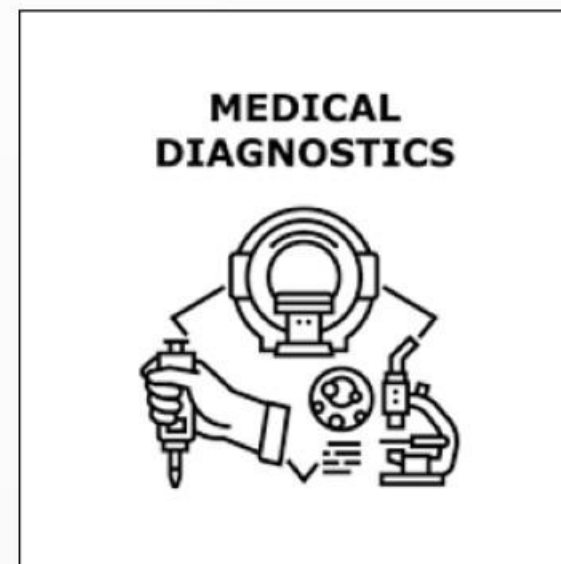


Presentation Overview

- **Problem Formulation & Clinical Motivation**
 - Diagnostic challenges and ML objectives.
- **Data Engineering & Preprocessing**
 - Dataset overview and pipeline details.
- **Model Architecture & Strategy**
 - Ensemble approach and model selection.
- **Evaluation & Interpretability**
 - Performance metrics, SHAP, and error analysis.
- **Real-Time Deployment**
 - Live demo of the "Health Symptom Analyzer" tool.
- **Conclusion & Future Scope**
 - Summary of results and roadmap for clinical integration.

Problem Formulation & Clinical Motivation

- **The Diagnostic Challenge:**
 - **High Ambiguity:** Non-unique Symptoms (e.g., Fever maps to >20 classes).
 - **Sparse Data:** Patients present limited information (3-5 symptoms out of 131 possibilities).
- **Machine Learning Objective:**
 - Build a **Multi-Class Classification Pipeline**.
 - Input: Binary Symptom Vectors ($x \in \{0,1\}^{131}$).
 - Output: Disease Probability Distribution ($y \in \{C_1, \dots, C_{41}\}$).
- **Optimization Goal:**
 - Maximize **Precision** and **Recall** to minimize false negatives (critical in healthcare).



Source: Medical diagnostics icon by Vecteezy

Data Engineering & Preprocessing

- **Dataset Overview:**

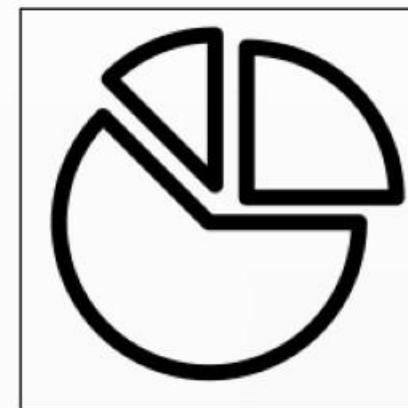
- **Source:** Synthetic clinical records of Symptom-Disease pairs.
- **Volume:** 4,920 Total Samples.
- **Target:** 41 Unique Disease Classes (Prognosis).

- **Feature Space:**

- **Input:** 131 Unique Symptoms (e.g., itching, nausea, high_fever).
- **Format:** One-Hot Encoded Binary Vectors (0 = Absent, 1 = Present).

- **Preprocessing Pipeline:**

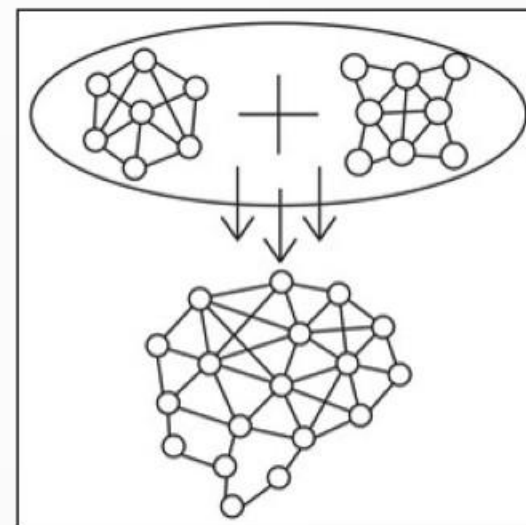
- **Data Cleaning:** Removal of duplicates and null values.
- **Label Encoding:** Transformed disease names into numerical classes (0-40).
- **Validation Strategy:** Stratified 80/20 Train-Test split to preserve class distribution.



Source: Split pie chart icon by The Noun Project

Model Architecture & Ensemble Strategy

- **Candidate Models (7 Evaluated):**
 - **Linear:** Logistic Regression, SVM (Linear Kernel).
 - **Tree-Based:** Decision Tree, Random Forest, XGBoost.
 - **Probabilistic/Instance:** Naive Bayes, KNN.
- **Selected Strategy: Soft Voting Ensemble**
 - **Mechanism:** Aggregates predicted probabilities from all base estimators.
 - **Goal:** Reduce variance and improve robustness against noise.
- **Configuration:**
 - **Random Forest:** Weight = 5 (Highest stability).
 - **Others:** Weight = 1 (Diversity).



Source: Ensemble learning methods icon by Shutterstock

Test Set Accuracy: 99.10% (Voting Ensemble)

- Qualitative Insights:

- [illegible]

Real-Time Deployment & User Interface

- **Tech Stack:**

- **Frontend:** Streamlit (Python-based reactive web framework).
- **Inference Engine:** joblib for low-latency serialization of the Ensemble Model.

- **Key Features:**

- **Dynamic Search:** Handles high-cardinality inputs (131+ symptoms) via searchable dropdowns.
- **Safety Protocols:** Integrated medical disclaimers and "insufficient data" warnings to ensure ethical AI usage.
- **User Experience:** Professional "Hospital Light" theme with glassmorphism for clarity.



The screenshot displays the 'HEALTH SYMPTOM ANALYZER' web application. It features a light blue background with a professional, clean design. At the top, the title 'HEALTH SYMPTOM ANALYZER' is prominently displayed in blue. Below it, the section 'DESCRIBE YOUR SYMPTOMS' is followed by a prompt: 'Please select up to 5 symptoms from the dropdowns below to receive a preliminary assessment.' There are five symptom selection dropdowns, each labeled 'Symptom 1' through 'Symptom 5'. The first three dropdowns are populated with 'itching', 'Skin Rash', and 'Nodal Skin Eruptions' respectively. The last two are set to 'Select a symptom...'. A large, rounded blue button labeled 'Analyze Health Condition' is positioned below the dropdowns. Underneath this button, a light blue box contains the 'ASSESSMENT RESULT: FUNGAL INFECTION'. At the bottom, a white box with a yellow warning icon contains a 'Medical Disclaimer' stating that the tool is a Machine Learning (ML) tool for preliminary assessment and should not be used as a substitute for professional medical advice.

HEALTH SYMPTOM ANALYZER

DESCRIBE YOUR SYMPTOMS

Please select up to 5 symptoms from the dropdowns below to receive a preliminary assessment.

Symptom 1
itching

Symptom 2
Skin Rash

Symptom 3
Nodal Skin Eruptions

Symptom 4
Select a symptom...

Symptom 5
Select a symptom...

Analyze Health Condition

ASSESSMENT RESULT: FUNGAL INFECTION

Medical Disclaimer: This Machine Learning (ML) tool analyzes symptoms to provide a preliminary assessment. While it is designed to assist in identifying potential conditions, it should not be solely relied upon. This result is not a substitute for professional medical advice; please consult a qualified doctor for a formal diagnosis and treatment.

Conclusion & Future Scope

- **Project Summary**
 - Engineered end-to-end ML pipeline.
 - **Soft Voting Ensemble** outperformed all baselines.
 - Achieved **99.2% Accuracy** with high stability.
 - Delivered an interpretable, real-time triage tool.
- **Future Roadmap**
 - **Severity Scoring:** Weigh symptoms (e.g., Mild vs. Severe).
 - **NLP Integration:** Extract symptoms from text using BioBERT.
 - **Scalability:** Dockerize for cloud deployment (AWS/Azure).

References

- [1] "Disease Prediction Using Machine Learning," Kaggle. [Online]. Available: <https://www.kaggle.com/datasets/kaushil268/disease-prediction-using-machine-learning>. [Accessed: Dec. 2, 2025]. (Note: This is the standard source for the 'Prototype.csv' dataset commonly used in this course).
- [2] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why Should I Trust You?: Explaining the Predictions of Any Classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, San Francisco, CA, USA, 2016, pp. 1135–1144. (Citation for LIME).
- [3] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems 30*, 2017, pp. 4765–4774. (Citation for SHAP).
- [4] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011. (Citation for the Scikit-learn library).
- [5] "Medical diagnostics icon vector illustration," Vecteezy. [Online]. Available: <https://www.vecteezy.com/vector-art/9905881-medical-diagnostics-icon-vector-illustration>. [Accessed: Dec. 2, 2025]. (Citation for your graphics).



Thank You!